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Q.P.

$$1. \quad \frac{dw}{dv} = f(v) - w + I_a \quad \frac{dw}{dv} = bv - w$$

$$f(v) = \begin{cases} -v & \text{for } v \leq 1 \\ \frac{1}{2}v - \frac{3}{2} & 1 \leq v \leq 5 \\ b-v & v \geq 5 \end{cases}$$

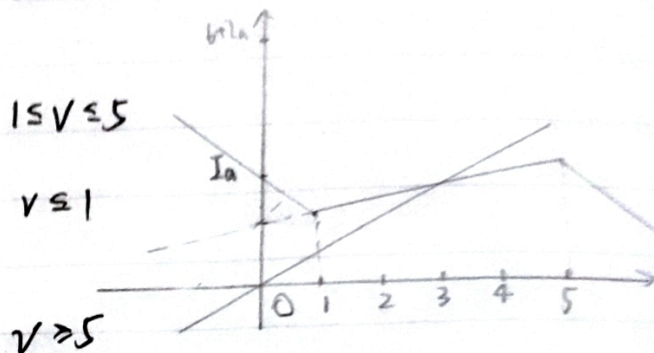
$$f(v) = w - I_a$$

$$\Rightarrow \begin{cases} 1 \leq v \leq 5 \\ \end{cases} \begin{cases} f(v) = \frac{1}{2}v - \frac{3}{2} = I_a + w \\ w = bv \end{cases} \Rightarrow \frac{1}{2}v - \frac{3}{2} = bv - I_a$$

$$\Rightarrow \begin{cases} w = bv \\ w = \frac{1}{2}v - \frac{3}{2} + I_a \end{cases} \quad 1 \leq v \leq 5$$

$$\begin{cases} w = bv \\ w = -v + I_a \end{cases} \quad v \leq 1$$

$$\begin{cases} w = bv \\ w = b - v + I_a \end{cases} \quad v \geq 5$$



When $I_1 < I_a < I_2$

$$\therefore b > \frac{1}{2}$$

$$5b > 1 + I_a, b < -1 + I_a$$

$$\Rightarrow \begin{cases} b > \frac{1}{2} \\ b+1 < I_a < 5b-1 \end{cases}$$

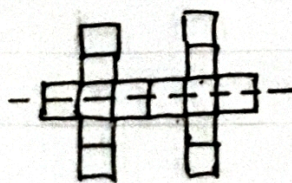
2.

Solve:

A

$$dI_1 = dm \cdot r^2$$

$$\Rightarrow dI_1 = b \times 2 \times r^2 dr$$



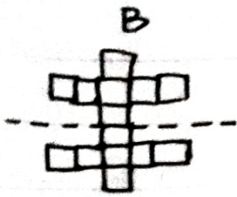
$$\Rightarrow I_1 = \int_{-1}^1 2r^2 dr = 4r^3 \Big|_{-1}^1 = 8$$

$$dI_2 = 2 \times 2 \times r^2 dr = 4r^2 dr$$

$$\Rightarrow I_2 = \int_{-1}^5 4r^2 dr = \frac{4}{3} r^3 \Big|_{-1}^5 = \frac{4}{3} \times (125 - 1) = \frac{496}{3}$$

$$\Rightarrow I_A = I_1 + 2I_2 = 338.67$$

I_B



$$dI_1 = 4 \times 2r^2 dr$$

$$\Rightarrow I_1 = \int_2^4 8r^2 dr = \left. \frac{8}{3} r^3 \right|_2^4 = \frac{8}{3} \times (64 - 8) = 149.3$$

$$I_2 = \int_{-6}^6 2r^2 dr = \left. \frac{2}{3} r^3 \right|_{-6}^6 = \frac{2}{3} \times (216 \times 2) = 288$$

$$\Rightarrow I_B = 2I_1 + I_2 = 298.6 + 288 = 586.6$$

$\Rightarrow B$ resist the bending better.